SVKM'S NMIMS
Mukesh Patel School of Technology Management & Engineering
A.Y. 2025 – 26

Course: Machine Learning
Project Report

Mall Customer Spending Analysis & Strategy Recommendation
Using Machine Learning & Game Theory

Program	B.Tech – Artificial Intelligence and Machine Learning
Semester	Semester IV (A.Y. 2025–26)
Project Title	Mall Customer Spending Analysis & Strategy Recommendation Using ML
Dataset	Mall_Customers.csv – 200 customers, 5 features
Project Members	**Roll No. Name**
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Contribution of each Project Member:

Roll No.	Name	Contribution
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GitHub Link of Project: https://github.com/M270706/MallCustomerSegmentation

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1. Introduction

🛍️ Mall Customer Spending: Predicting Behavior for Smarter Marketing

The Problem We Solved Malls collect a massive amount of data on us, but they often struggle to actually *use* it to personalize our shopping experience. Instead of guessing, what if a mall knew exactly who to target with a discount? This project uses machine learning to figure out how customers spend their money and pairs that with a game-theory model to recommend the smartest marketing move. The ultimate goal? Help the mall maximize its revenue while giving shoppers offers they actually care about.

How We Did It

- **Predicting the "Spending Score":** First, we wanted to see if we could guess a shopper's exact spending score (on a scale of 1 to 100). We trained two machine learning models (Linear Regression and a Decision Tree Regressor) to predict this number using just a customer's age, annual income, and gender.
- **Grouping the Shoppers:** Next, we kept things simple by categorizing people into three distinct buckets: Low (0–39), Medium (40–69), and High (70–100) spenders. Using a Decision Tree Classifier, our model was able to accurately sort 80% of the customers into the right group.
- **The "Discount" Dilemma:** Who actually *needs* a discount to make a purchase? We designed a payoff matrix inspired by game theory to answer this. The system looks at a customer's spending segment and recommends either "Discount" or "No Discount"—ensuring the mall doesn't waste money subsidizing purchases that would have happened anyway.
- **Bringing It to Life:** We didn't want this to just be a bunch of code. We built an interactive web dashboard using Streamlit so anyone can easily visualize the customer segments, explore the data, and see the recommended strategies in real-time.

A Look at the Shoppers (The Data) All of this was built using a perfectly clean dataset of 200 mall customers (zero missing values, zero duplicates!). Here is a quick snapshot of who was walking around the mall:

- **The Mix:** 112 women and 88 men.
- **Age:** Shoppers ranged from 18 to 70 years old, with the average shopper being about 39.
- **Income:** Annual incomes were incredibly diverse, stretching from $15k all the way to $137k (averaging about $60k).
- **Spending Habits:** Their assigned spending scores spanned almost the entire 1–100 scale (averaging right in the middle at 50.2).

Note: All the coding, testing, and modeling for this project was done in Python using Google Colab.

2. Literature Survey

📚 Standing on the Shoulders of Giants (Past Research)

Figuring out how and why customers shop isn't exactly a new idea. To build our model, we looked back at decades of retail and mathematical research to see what actually works:

- **The Classic Grouping Method (1967):** The idea of grouping similar shoppers together actually started way back in the late 60s with a technique called *k-means clustering* (introduced by J. MacQueen). It's been the gold standard for retail segmentation ever since.
- **Follow the Money (1994):** The famous RFM analysis (Recency, Frequency, Monetary) proved a very simple truth: the best way to guess what a customer will do next is to look closely at their past spending habits.
- **The Machine Learning Upgrade (2010s):** Over the last decade, researchers figured out that feeding rich customer data into supervised machine learning models results in much sharper, more accurate spending predictions than old-school statistics.

- **The Math of Making Deals (1950):** We even pulled from John Nash's foundational Game Theory. For decades, businesses have used his frameworks to figure out the perfect pricing and discount strategies to outsmart the competition.
- **Decision Trees (2015–2020):** Recent studies confirmed that "Decision Trees" are fantastic for sorting customers because they hit the sweet spot: they are highly accurate, but still easy for humans to read and understand.

✨ What Makes This Project Stand Out?

Even with all that history, a lot of academic research stays stuck in theory. We wanted to bring something fresh and highly practical to the table:

- **The Best of Both Worlds:** Most data projects either try to predict an exact number (regression) *or* group people into categories (classification). We decided to run both pipelines on the exact same dataset to get a much richer, 360-degree view of the shoppers.
- **Math Meets Marketing:** It's surprisingly rare to see a project that takes machine learning predictions and plugs them directly into a formal Game Theory payoff matrix. We didn't just want to tell the mall *who* their customers were; we wanted to tell them exactly *what to do* with them to maximize profits.
- **Ready to Use Today:** A lot of brilliant data science gets buried in messy, offline code files. We brought ours to life. By deploying a live web dashboard via Streamlit, mall management can jump in, filter the data, and explore these insights in real-time without needing to know a single line of code.

3. Methodology

🗺️ The Step-by-Step Pipeline: From Raw Data to Real Strategy

Here is exactly how we turned a basic spreadsheet of mall customers into an interactive, predictive business tool:

1. Getting to Know the Data (Loading & Exploration)

We started by importing the Mall_Customers.csv file into Python. A quick health check confirmed we had a pristine dataset: 200 shoppers, 5 key details each, and

absolutely zero missing pieces of information. We also ran some quick summary statistics to get a feel for the averages.

2. Cleaning Things Up (Preprocessing)

Machine learning models love numbers, not words. So, we converted the text-based "Gender" column into a simple numerical format (Gender_Male, where 1 = Male and 0 = Female). We also tossed out the CustomerID—after all, a random ID number doesn't tell us anything about how someone shops!

3. Finding the Odd Ones Out (Outlier Detection)

Were there any wildly unusual shoppers skewing our data? We used a classic statistical trick (the IQR method) and visual box plots to investigate. We found that Age and Spending Scores were very well-balanced, but we did spot a few "whales"—customers with unusually high annual incomes.

4. Setting the Stage (Train/Test Split)

We defined our "clues" (Features: Age, Income, Gender) and our ultimate goal (Target: Spending Score). To make sure our models were actually learning and not just memorizing, we split the data: 80% was used to train the AI, and 20% was kept completely hidden to test its accuracy later.

5. Predicting the Exact Score (Regression)

First, we tried to predict a customer's *exact* spending score on a scale of 1 to 100. We put two models head-to-head—a Decision Tree Regressor and a Linear Regression model—and compared their error rates (MAE, MSE, RMSE) to see which one was the most accurate guesser.

6. Creating Customer Buckets (Categorization)

Because predicting exact numbers is tough, we also grouped the shoppers into three simple, actionable buckets:

- **Low Spenders:** Scores 0–39 (59 customers)
- **Medium Spenders:** Scores 40–69 (87 customers)
- **High Spenders:** Scores 70–100 (54 customers)

7. Predicting the Groups (Classification)

Next, we trained a Decision Tree Classifier to predict exactly which of those three buckets a new customer would fall into. We evaluated its performance using a confusion matrix to see exactly where the model got it right and where it got confused.

8. The Business Strategy (Game Theory Payoff)

Here is where the data makes money. We built a "payoff matrix" to determine the most profitable action for each customer type.

- **The Strategy:** The 54 High Spenders were assigned a "No Discount" strategy (why sacrifice profit margins on guaranteed buyers?). The remaining 146 Low and Medium Spenders were assigned a "Discount" strategy to tempt them into engaging.

9. Bringing It to Life (Streamlit Dashboard)

Finally, we didn't just leave this as a bunch of code. We built a live, interactive web dashboard using Streamlit (hosted via ngrok). Now, anyone can view the raw data, play with interactive scatter plots, see the strategy pie charts, and filter the customers with the click of a button.

4. Result and Discussion

4.1 Regression Model Performance

Both models were trained on the three-feature set (Age, Annual Income (k$), Gender_Male) and evaluated on the held-out 20% test set (random_state=50). Results are presented in Table 4.1.

Table 4.1 – Regression Model Evaluation Results

Model	MAE	MSE	RMSE	R^2	Verdict
Decision Tree Regressor	19.32	967.17	31.10	-0.49	Overfit / Weak
Linear Regression ★ (Better)	**20.46**	**598.52**	**24.46**	**0.08**	**Moderate**

Linear Regression achieved a lower MSE (598.52 vs 967.17) and a positive R^2 (0.08) compared to the Decision Tree Regressor ($R^2 = -0.49$), indicating that LR generalises better on unseen data. The negative R^2 for the Decision Tree reveals severe overfitting — the model memorised the training data rather than learning

the underlying pattern. Both models reflect that spending score is highly non-linear and influenced by factors not captured in this three-feature set.

4.2 Customer Spending Category Distribution

Customers were grouped into three spending tiers using pd.cut with bins=[0, 40, 70, 101] (right=False). Medium spenders form the largest group (87), followed by Low (59) and High (54).

Table 4.2 – Spending Category Distribution

Category	Score Range	Count
Low	0 – 39	59
Medium	40 – 69	87
High	70 – 100	54

4.3 Decision Tree Classifier Performance

The Decision Tree Classifier was evaluated on a stratified 20% test set (40 samples). The model achieved an overall accuracy of 80%. Table 4.3 shows per-class metrics.

Table 4.3 – Classification Report

Class	Precision	Recall	F1-Score	Support
High	0.62	0.73	0.67	11
Low	0.90	0.75	0.82	12
Medium	0.88	0.88	0.88	17
Overall Accuracy			**0.80**	40

Medium and Low spending classes were classified with high precision (0.88 and 0.90 respectively). The High class had the lowest precision (0.62) but best recall (0.73), indicating the model catches most high spenders but occasionally misclassifies medium-spenders as high.

4.4 Game Theory Strategy Recommendation

Using the payoff matrix, 54 high-spending customers (27%) received a 'No

Discount' recommendation for maximum revenue retention, while 146 low/ medium spenders (73%) were recommended a 'Discount' strategy to stimulate engagement.

Table 4.4 – Payoff Matrix

Strategy	Customer Accepts	Customer Rejects	Applied To
No Discount	+80	0	High spenders (n=54)
Discount	+50	-10	Low/Medium spenders (n=146)

5. Conclusion

🏆 The Big Wins (Key Takeaways)

This project proved that you don't need a massive tech team to turn raw retail data into smart business decisions. Here is what we learned:

- **Simple Math Won:** When trying to predict exact scores, the simplest model (Linear Regression) actually generalized better to new data than the complex AI (Decision Tree Regressor). It proves that when dealing with small datasets, keeping the math simple is often the smartest choice.
- **80% Accuracy on Segments:** Our Decision Tree Classifier was able to accurately bucket 80% of the shoppers into the right spending category. It was especially great at identifying the "Medium" spenders—the core demographic of the mall.
- **Strategy Over Statistics:** The Game Theory payoff matrix is where the magic happened. Instead of just handing the mall a spreadsheet of predictions, we gave them a concrete, mathematically backed action plan (Discount vs. No Discount) for every single shopper.
- **Made for Humans:** By deploying the Streamlit dashboard, we took complex data science and made it accessible. A mall manager without any coding experience can now click around, filter demographics, and see their marketing strategy laid out visually.

🚧 Where We Hit Roadblocks (Limitations)

Every good data project has its limits, and ours is no exception:

- **Small Sample Size:** We only had data for 200 people. While that's enough to spot general trends, it's incredibly hard for regression models to find bulletproof patterns with so few examples (which resulted in lower R^2 scores).
- **Missing the "What" and "When":** We knew the shoppers' ages and incomes, but we didn't know *what* they were buying or *how often* they visited. Adding purchase history and visit frequency would make these predictions drastically more accurate.
- **Theoretical Math:** The payoff matrix values (+50, -10, etc.) were manually set based on logical assumptions. In the real world, these numbers would need to be dynamically calibrated using actual mall revenue and profit margins.

🚀 What's Next? (Future Enhancements)

If we were to take this project to the next level, here is exactly what we would do:

- **Mix and Match Models:** We want to combine this with "unsupervised" learning (like k-means clustering) to create even deeper, highly specific customer personas (e.g., "Young High-Income Tech Buyers" vs "Older Bargain Hunters").
- **Bring in the Heavy AI:** If we can get a dataset with thousands of shoppers, we could train deep neural networks to catch the tiny, non-linear spending habits that basic models miss.
- **Live Cash Register Data:** Imagine plugging this model directly into the mall's point-of-sale (POS) systems. The dashboard would update in real-time as people swipe their credit cards!
- **Hyper-Personalization:** Instead of just recommending a generic "Discount," the ultimate goal is to recommend *exactly* what to discount (e.g., "Give Customer #42 a 15% coupon for shoes") to maximize their likelihood of buying.

6. References

⚙️ Under the Hood: The Tools & Data

The Dataset

- **Source:** UCI / Kaggle (Customer Segmentation Dataset).
- **The Blueprint:** Mall_Customers.csv contains 200 individual customer records tracked across 5 key features (CustomerID, Gender, Age, Annual Income, and Spending Score). It provided the perfect, clean sandbox for our predictive models.

The Tech Stack (Tools & Libraries)

We didn't just build a model; we built a full pipeline. Here is the technology that made it happen:

- **The Heavy Lifters (Python & Pandas):** Used for loading the data, cleaning it up, and engineering new features (like transforming gender into binary code and hunting down extreme outliers using the IQR method).
- **The Brains (Scikit-learn):** The core machine learning engine. We relied on this library to power our Linear Regression, Decision Tree Regressor, and Decision Tree Classifier, as well as to split our data and grade our models' accuracy.
- **The Visuals (Matplotlib & Seaborn):** Used to turn raw numbers into beautiful, understandable graphics—from scatter plots of income versus spending to heatmaps showing where our models got confused.
- **The Stage (Streamlit & pyngrok):** We used Streamlit to build the interactive web dashboard, and pyngrok to broadcast it live to the public via a secure URL.

📚 Academic & Strategic Foundations

This project stands on the shoulders of decades of mathematical and retail research. Here are the foundational papers and texts that guided our approach:

Customer Segmentation & Strategy

- **MacQueen, J. (1967).** *Some Methods for Classification and Analysis of Multivariate Observations.* (The birth of modern clustering and segmentation).
  - Semantic Scholar | Direct PDF (via SciSpace)
- **Hughes, A.M. (1994).** *Strategic Database Marketing.* (Proving that past spending behavior predicts future value).
  - Google Books | Internet Archive
- **Nash, J. (1950).** *Equilibrium Points in n-Person Games.* (The Game Theory mechanics used to design our discount payoff matrix).
  - PNAS Official Publication | **DOI:** 10.1073/pnas.36.1.48

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- **Breiman, L. (2001).** *Random Forests.* (Foundational reading for advanced Decision Tree architecture).
  - UC Berkeley Archival PDF | Springer Publication
- **Romero, C. & Ventura, S. (2010).** *Educational Data Mining: A Review of the State-of-the-Art.* * IEEE Xplore | **DOI:** 10.1109/TSMCC.2010.2053532
- **Fernandes, E. et al. (2019).** *Educational Data Mining: Predictive Analysis of Academic Performance.* * ScienceDirect | **DOI:** 10.1016/j.jbusres.2018.02.012